

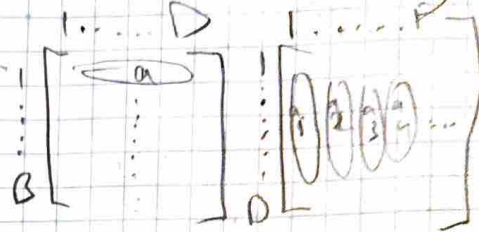
Q) Int8 matmul.

We want to do $X[B,D] \cdot Y[D,F] \rightarrow Z[B,F]$
with int8 precision

1] To load X , we need to load BD bytes (BD params and 1 byte param)
To load Y , " " " " DF " " " "
To store Z , we need to store BF bytes (BF " " " ") ✓

2] $X @ Y$ is BF dot products
for each row, take F dot products.

Each dot product is a dotting



two D -dim vectors, which was previously established to take

N multiplications, $N-1$ additions, for $2N-1$ ops.

So, total ops = $BF(2D-1) = 2BDF - BF$

3] This is $\frac{BF(2D-1)}{BD+DF+BF}$. If we assume B is small, then

we have $\frac{BF(2D-1)}{DF} = B \left(\frac{(2D-1)(F)}{DF} \right)$, and in the limit $\rightarrow 1$,

we have $\frac{2D-1}{D} = \frac{2D}{D} = 2$, so

Intensity = ~~$2B$~~ $= 2!$

4] Reference

Arithmetic Intensity,

$\frac{3.94e14}{8.2e11} = \frac{3.94e3}{8.2} = 480$

Realistic estimate,

where $480 \approx 2!$

$\Rightarrow B \geq 240$ ← The point at which it becomes complex bound

$$T_{\text{math}} = \frac{BF(20-1)}{3.44e14}, \quad T_{\text{comm}} = \frac{BD+DF+BF}{5.2e11}$$

Lower bound is $\max(T_{\text{math}}, T_{\text{comm}})$ Upper bound is $T_{\text{math}} + T_{\text{comm}}$

Q2] Store weights in low precision keeps ops + calculations 2nd part

$$b16[B,D] @ \text{int8}[D,F] \rightarrow b16[B,F]$$

$$T_{\text{comm}} = T_{\text{math}} + T_{\text{real}} = 2BF + 2BD + DF$$

$$T_{\text{math}} = BF(20-1) \text{ which is } \text{about } 20BF$$

Compare bound when $\text{arithmetic intensity} > \text{data intensity}$

$$\text{which is when } \frac{T_{\text{math}}}{T_{\text{comm}}} > \frac{1.97e14}{5.2e11} = 240$$

$$\Rightarrow \frac{20BF}{2BF+2BD+DF} > 240 \text{ since } BF \ll DF$$

$$\text{so, } \Rightarrow \frac{20BF}{DF}, \text{ so when } B \geq 20, \text{ we have } \text{arithmetic bound}$$

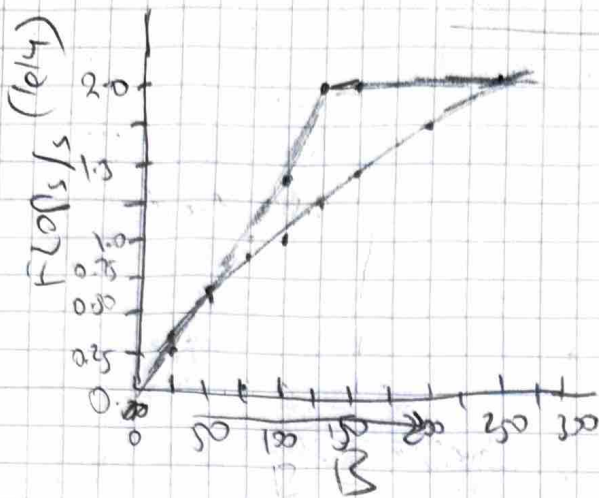
Q3) Profile Plot,

for $F=D=4096$, we load $2(4096)B + 4096^2$ bytes

and write $2B(4096)$ bytes.

$$= 16384B + 16,777,216 \text{ bytes}$$

4096



Operation is

$$BF(D-1) =$$

$$4096B(8191) = 33,550,336 \text{ op Flops}$$

$$\text{which takes } \frac{33,550,336}{1.97e4} = 1.70e-7 \text{ seconds}$$

and memory takes

$$\frac{16,777,216}{8.21e4} = 2.04e-5 \text{ seconds}$$

for

x Flops, y writes

as ~~plot~~

$$\text{total Flops} = f$$

$$\text{total mem. mes.} = m$$

$$\frac{f}{m} = \text{Intensity} \rightarrow \text{Intensity, mem.}$$

$$\frac{\text{Flops}}{\text{bytes}} = \text{accelerating}$$

$$\text{bytes/s} \rightarrow 2 \cdot \text{bytes/s}$$

$$\frac{f}{m} \cdot \frac{\text{Flops}}{\text{bytes}} \times \frac{\text{bytes}}{s} = \frac{\text{Flops}}{s}$$

$$(04) \text{Mat}^8 [B, D] \circ \text{Mat}^8 [B, D, F] \rightarrow \text{Mat}^8 [B, F]$$

$$t_{\text{comm}} = \underset{\text{load}}{BD} + \underset{\text{load}}{BDF} + \underset{\text{write}}{BF}$$

$$t_{\text{math}} = 2BDF$$

a different matrix per
batch element does all
change FLOPs,

$$\frac{t_{\text{math}}}{t_{\text{comm}}} = \frac{2BDF}{BD + BDF + BF} = \frac{2BDF}{B(D + DF + F)} = \frac{2DF}{D + DF + F} = \text{if } DF \gg D + F \text{ (reasonable?)}$$

then this is just $\boxed{2}$
pretty terrible

the R. Kishy D is a constant 2, but!

(05). We have $1.979 \text{ teraFLOPs/s}$ and 3.35 TB/s
 $= 1.979 \times 10^{12} \text{ FLOPs/s}$

$$= (3.35)(1024)(1024)(1024)(1024) \text{ bytes/sec}$$

$$= 3.683 \times 10^{12} \text{ bytes/sec}$$

$$\text{Accelerator} = \text{Intensity (accelerator)} = \frac{1.979 \times 10^{12}}{3.683 \times 10^{12}} = 0.537$$

$$\text{Intensity (matrix)} \approx B, \text{ so } B \approx 1$$

We have 1.979 teraFLOPs

$$= 1.979 \times 10^{15} \text{ FLOPs}$$

and 3.35 TB/s

$$= 3.683 \times 10^{12} \text{ bytes}$$

Then, Intensity of accelerator = $\frac{1.979 \times 10^{15}}{3.683 \times 10^{12}} = 537$

so, $B \geq 538$ for matrix to be compute bound



but what speaks, we have $\frac{1.979 \times 10^{15}}{3.683 \times 10^{12}} = 268.5$

so we need $B \geq 269$